

Lecture 13: Backward Stability

Math 6610

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Today's Goals

- Define backward stability: $\tilde{F}(x) = F(\tilde{x})$ with $\tilde{x}$ near x
- Prove that floating-point subtraction is backward stable
- See an algorithm $(x + 1)$ that is stable but not backward stable
- State the accuracy theorem: relative error $= O(\kappa \epsilon_{\text{mach}})$

Recall: Stability (Lecture 12)

Definition (Stable). The algorithm $\tilde{F} : X \rightarrow Y$ for the problem F is *stable* if for each $x \in X$, $x \neq 0$, there exists $\tilde{x} \in X$ such that

$$(1) \quad \frac{\|\tilde{F}(x) - F(\tilde{x})\|}{\|F(\tilde{x})\|} = O(\epsilon_{\text{mach}}),$$

$$(2) \quad \frac{\|\tilde{x} - x\|}{\|x\|} = O(\epsilon_{\text{mach}}),$$

with constants in $O(\epsilon_{\text{mach}})$ that do not depend on x . (In (1), read the quotient as 0 if $\tilde{F}(x) = F(\tilde{x}) = 0$.)

Remark (Meaning). A stable algorithm gives nearly the right solution to nearly the right problem.

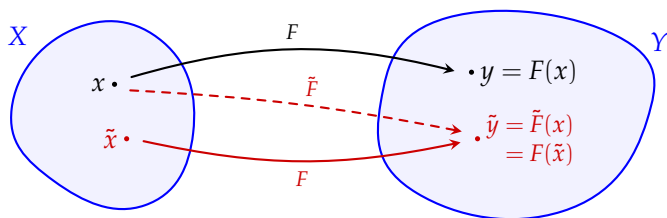
Backward Stability: Definition

Definition (Backward stability). The algorithm $\tilde{F} : X \rightarrow Y$ for the problem $F : X \rightarrow Y$ is *backward stable* if for each $x \in X$, $x \neq 0$, there exists $\tilde{x} \in X$ such that

$$\tilde{F}(x) = F(\tilde{x}) \quad \text{and} \quad \frac{\|\tilde{x} - x\|}{\|x\|} = O(\epsilon_{\text{mach}}),$$

with a constant in $O(\epsilon_{\text{mach}})$ that does not depend on x (Lecture 12).

Remark (Meaning). A backward stable algorithm gives the right solution to nearly the right problem.



Backward Stable Implies Stable

Proposition. Every backward stable algorithm is stable.

Proof. Take the $\tilde{x}$ from backward stability. Condition (2) of stability (Lecture 12) is the same condition. For condition (1), $\tilde{F}(x) - F(\tilde{x}) = 0$, so the relative difference is $0 = O(\epsilon_{\text{mach}})$. ■

Remark. The converse is false: the example $x + 1$ below is stable but not backward stable.

Subtraction Is Backward Stable

Proposition. Let $X = \mathbb{R}^2$ with a p -norm ($p = 1, 2, \infty$), $Y = \mathbb{R}$, and

$$F(x) = x_1 - x_2, \quad \tilde{F}(x) = \text{fl}(x_1) \ominus \text{fl}(x_2), \quad x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}.$$

Assume no overflow or underflow occurs. Then there is $\tilde{x} \in \mathbb{R}^2$ with

$$\tilde{F}(x) = F(\tilde{x}), \quad |\tilde{x}_i - x_i| \leq (\epsilon_{\text{mach}} + \tfrac{1}{4}\epsilon_{\text{mach}}^2) |x_i|, \quad i = 1, 2.$$

In particular $\tilde{F}$ is backward stable.

Subtraction Is Backward Stable: Proof (1/2)

Proof. 1. Write what is computed. One factor $(1 + \epsilon_i)$ per rounding: two for storing x_1, x_2 (Lecture 10), one for $\ominus$ (Lecture 11), with $|\epsilon_1|, |\epsilon_2|, |\epsilon_3| \leq \frac{1}{2} \epsilon_{\text{mach}}$:

$$\tilde{F}(x) = [x_1(1 + \epsilon_1) - x_2(1 + \epsilon_2)](1 + \epsilon_3).$$

2. Move the errors onto the data. Multiply out:

$$\tilde{F}(x) = \underbrace{x_1(1 + \epsilon_1)(1 + \epsilon_3)}_{=: \tilde{x}_1} - \underbrace{x_2(1 + \epsilon_2)(1 + \epsilon_3)}_{=: \tilde{x}_2} = F(\tilde{x}).$$

(continued on the next slide)

Subtraction Is Backward Stable: Proof (2/2)

Proof (continued). 3. *Size of each change.* For $i = 1$ (and in the same way for $i = 2$, with ϵ_2 in place of ϵ_1):

$$\begin{aligned} |\tilde{x}_1 - x_1| &= |x_1| |(1 + \epsilon_1)(1 + \epsilon_3) - 1| = |x_1| |\epsilon_1 + \epsilon_3 + \epsilon_1 \epsilon_3| \\ &\leq |x_1| (|\epsilon_1| + |\epsilon_3| + |\epsilon_1| |\epsilon_3|) \leq (\epsilon_{\text{mach}} + \tfrac{1}{4} \epsilon_{\text{mach}}^2) |x_1|. \end{aligned}$$

4. *From entries to norms.* Let $\delta = \epsilon_{\text{mach}} + \frac{1}{4} \epsilon_{\text{mach}}^2$. In a p -norm, a vector whose entries are bounded by $\delta |x_i|$ has norm at most $\delta \|x\|$, so

$$\frac{\|\tilde{x} - x\|}{\|x\|} \leq \epsilon_{\text{mach}} + \tfrac{1}{4} \epsilon_{\text{mach}}^2 \leq 2 \epsilon_{\text{mach}} \quad (\epsilon_{\text{mach}} \leq 1).$$

The constant 2 does not depend on x . ■

Remark. The same argument shows that $\oplus$, $\otimes$, $\oslash$ are backward stable (for $\otimes$, $\oslash$ put $(1 + \epsilon_3)$ on x_1 only).

Adding 1: Not Backward Stable (1/2)

Example ($F(x) = x + 1$, $X = Y = \mathbb{R}$). The algorithm, with $|\epsilon_1|, |\epsilon_2| \leq \frac{1}{2} \epsilon_{\text{mach}}$:

$$\tilde{F}(x) = \text{fl}(x) \oplus 1 = [x(1 + \epsilon_1) + 1](1 + \epsilon_2).$$

F is one-to-one, so $F(\tilde{x}) = \tilde{F}(x)$ has only one solution, $\tilde{x} = \tilde{F}(x) - 1$:

$$\tilde{x} = x(1 + \epsilon_1)(1 + \epsilon_2) + \epsilon_2.$$

Its relative change:

$$\left| \frac{\tilde{x} - x}{x} \right| = \left| \epsilon_1 + \epsilon_2 + \epsilon_1 \epsilon_2 + \frac{\epsilon_2}{x} \right|.$$

The term ϵ_2/x is not bounded by a constant times ϵ_{mach} when x is near 0 and $\epsilon_2 \neq 0$.

Adding 1: Not Backward Stable (2/2)

Example (A specific input). Take $x = \epsilon_{\text{mach}}^2 = 2^{-104}$. Then $\text{fl}(x) = x$, and $1 + 2^{-104}$ lies less than half a spacing ($\frac{1}{2}\epsilon_{\text{mach}}$) above 1, so it rounds to 1:

$$\tilde{F}(x) = 1, \quad \tilde{x} = \tilde{F}(x) - 1 = 0, \quad \left| \frac{\tilde{x} - x}{x} \right| = 1.$$

Remark (Conclusion). For every precision ϵ_{mach} , the input $x = \epsilon_{\text{mach}}^2$ gives relative change 1. A bound $1 \leq C \epsilon_{\text{mach}}$ fails for $\epsilon_{\text{mach}} < 1/C$, whatever the constant C . So $\tilde{F}$ is not backward stable.

Remark (Reason). The rounding ϵ_2 of the addition must be moved onto x . The constant 1 is part of the problem and cannot take any of it. Relative to a small x , the added ϵ_2 is large.

Adding 1: Stable

Proposition. The algorithm $\tilde{F}(x) = \text{fl}(x) \oplus 1$ for $F(x) = x + 1$ is stable.

Proof. 1. *Choice of $\tilde{x}$.* Let $\tilde{x} := \text{fl}(x) = x(1 + \epsilon_1)$. Then

$$|\tilde{x} - x| = |\epsilon_1| |x| \leq \frac{1}{2} \epsilon_{\text{mach}} |x|.$$

2. *Output.* With this $\tilde{x}$, the algorithm computes

$$\tilde{F}(x) = (\tilde{x} + 1)(1 + \epsilon_2) = F(\tilde{x})(1 + \epsilon_2), \quad \text{so} \quad \tilde{F}(x) - F(\tilde{x}) = \epsilon_2 F(\tilde{x}).$$

Hence $|\tilde{F}(x) - F(\tilde{x})| \leq \frac{1}{2} \epsilon_{\text{mach}} |F(\tilde{x})|$ (both sides are 0 if $F(\tilde{x}) = 0$). Both conditions of stability hold with constant $\frac{1}{2}$, independent of x . ■

Accuracy of Backward Stable Algorithms

Theorem. Let the problem $F : X \rightarrow Y$ have condition number $\kappa(x)$ at $x \in X$ (Lecture 9), and let $\tilde{F} : X \rightarrow Y$ be a backward stable algorithm for F . Then, for all $x \in X$ with $x \neq 0$ and $F(x) \neq 0$,

$$\frac{\|\tilde{y} - y\|}{\|y\|} = \frac{\|\tilde{F}(x) - F(x)\|}{\|F(x)\|} = O(\kappa(x) \epsilon_{\text{mach}}),$$

i.e. bounded by $C \kappa(x) \epsilon_{\text{mach}}$ for small ϵ_{mach} , with C independent of x . (Proof: Lecture 14.)

Remark. If $\kappa(x)$ is large, even a backward stable algorithm can give a large relative error.

Reading the Accuracy Theorem

The theorem separates the two sources of error:

$$\underbrace{\frac{\|\tilde{y} - y\|}{\|y\|}}_{\text{error of the result}} = O\left(\underbrace{\kappa(x)}_{\text{problem}} \cdot \underbrace{\epsilon_{\text{mach}}}_{\text{algorithm}}\right).$$

- Backward stable algorithm, $\kappa(x)$ of moderate size $\Rightarrow$ about 16 correct digits.
- $\kappa(x) = 10^{10} \Rightarrow$ error up to about $10^{10} \cdot 10^{-16} = 10^{-6}$: about 6 correct digits. In the worst case no algorithm does better, since rounding the input alone can cause this error.
- An algorithm that loses many digits on a well-conditioned problem is not backward stable and should be replaced (e.g. LU without pivoting).

Looking Ahead

- **Next lecture:** Backward Stability; Vector and Matrix Norms